

Screen A: database design

Three one-line prompts and one four-question form, and Claude Code produced a 216-line schema design, eight recorded decisions, a new requirement and a revision of the basic design — for the production-order create/edit screen.

Presented alongside `ScreenA_DB_edited.mp4` (3 min 59 s). Step numbers on the slides match the captions in the video.
Recorded 2026-09-18 · Claude Code (Opus 5) in the JetBrains Rider terminal, auto mode on.

`0002-production-order-schema.md`

`decisions.md`

`brief.md`

`BD-001-...-create-edit.md`

`plan.md`

`status.md`

`evidence.md`

Eleven minutes of recording, four and a half of agent work

The human wrote three prompts and answered one form. Everything else was the agent working through the harness.

3

prompts typed

each one line long

4

questions answered

in a single form

4 m 19 s

agent working time

2 m 15 s + 13 s + 1 m 51 s

8

decisions recorded

DEC-013 to DEC-020

The human

- Approved plan revision 1 and asked for the database design.
- Asked whether any open questions were left — the pivot of the session.
- Chose an answer for each of four questions. One was free text, and on the database-login split they **overrode the recommendation**.

The agent, guided by the harness

- Read the database skill, rules and template — and the persistence code — before writing anything.
- Decided three schema-level technical points itself, each recorded with its options and rationale.
- Corrected its own draft twice, unprompted.
- Reported what the user's choice would cost: a plan revision touching WI-001's Compose setup.

12 steps, one video

Each step is a clip in the edited video. **Amber: the human acts.** **Teal: the agent works and produces output.**

STEP 0100:00 Prompt 1 Approve the plan, start DB design	STEP 0200:12 Reads first Skill, rules, template, then the code	STEP 0300:32 DB-002 written 216 lines of schema design	STEP 0400:46 Self-correction Rewrites its own operational note
STEP 0500:59 Plan · status · evidence Approval written into the work item	STEP 0601:15 Turn 1 summary DEC-013–DEC-015 taken, DEC-016 asked	STEP 0701:33 Prompt 2 "Are there any open questions left?"	STEP 0801:44 Five open items Three of them previously unrecorded
STEP 0902:02 Prompt 3 "Let's answer all of them"	STEP 1002:11 Decision form Four tabs, then review and submit	STEP 1102:50 Recording the answers DEC-016–DEC-020 and REQ-019	STEP 1203:32 Review the schema decisions.md and DB-002

Times are positions in `ScreenA_DB_edited.mp4`. Long stretches of agent work are jump-cut to their result; nothing is sped up.

One line approves the plan and starts the schema

THE PROMPT

Approved the plan;
let's move on to DB design

Claude Code had pre-filled a suggested reply — *"plan approved, let work on the database design next"*. The user cleared it and typed their own. The prompt names no file, no template and no schema.

OUTPUT

No artifact yet. The approval itself is the output: it unlocks milestone 4 of plan revision 1, which the previous session stopped at.

```
... and the screen is now blank.

A few technical choices remain for the next design steps. They don't change how the screen behaves:
- Database design (DB-002):
  - How to generate the number sequence that restarts each year.
  - What happens if a year goes past 99,999 orders.
  - How the system detects that someone else changed the order.
- Detailed design (DD-001): whether the new save endpoints need protection against cross-site request forgery (CSRF). The first pass concluded no extra token was needed; I'll c

Plan revision 1 still needs your review. Once you approve it, I'll start DB-002.

* Worked for 1m 37s · done 11:02 AM

} approved the plan, let move on to DB design
>> auto mode on (shift+tab to cycle)
```

Video 00:06. The prompt replacing Claude Code's suggestion.

Read the rules, then read the code

WHAT CLAUDE READ	WHAT IT GOT FROM IT
database-design skill, ai/rules/database.md	The structure DB-002 must follow, and the bar: least privilege, justify every index, state migration impact
ai/templates/ database-design.md	The section order, including an <i>Open decisions</i> table
DB-001, database README	The conventions the identity schema already set
AppDbContext.cs, EF migrations	snake_case naming, uuid keys — what is actually built, not what the docs claim

OUTPUT

"Plan approved, so I'm recording that and starting DB-002. I've checked the existing persistence setup (snake_case naming convention, uuid keys, EF migrations) so DB-002 can follow it."



Video 00:18. Read-only commands, collapsed by Claude Code.

216 lines of schema, then Claude fixes its own sentence

STEP 3 — DOCS/EN/DATABASE/0002-PRODUCTION-ORDER-SCHEMA.MD

TABLE	PURPOSE
products	Minimal product reference data, seeded, no catalog UI
production_orders	One row per production order created on SCR-001
production_order_number_counters	Last issued sequence per year, to generate PO-YYYY-NNNNN

Plus constraints and indexes with a reason each, an API-field mapping for DD-001 to confirm, and a rollback command with the warning that rolling back after use deletes all order data.

```
192 -- Rollback plan: "Delete of database update InitialIdentitySchema --project src/backend/ProductionManagementAI.Infrastructure --startup-project src/backend/ProductionManagementAI.Api" drops the three tables
193 -- Operational note: the generated 'order_number' column is written by raw SQL in the migration (EF Core 'HasComputedColumnSql(..., stored: true)'). The expression uses explicit '::text' casts because PostgreSQL requires generation expressions to be immutable.
194 -- Operational note: the generated 'order_number' column is defined with EF Core 'HasComputedColumnSql(..., stored: true)'. The expression uses explicit '::text' casts because PostgreSQL requires generation expressions to be immutable.
195 -- Open decisions
196
```

Video 00:52. Step 4: Claude rewrites line 193 of its own draft.

STEP 4 — SELF-CORRECTION

"...the generated `order_number` column is written by raw SQL in the migration" became "...is defined with EF Core `HasComputedColumnSql(..., stored: true)`". Nobody asked for it.

The approval is a file change, not a memory

- **plan.md** — "Review status: submitted for review" becomes **approved**; the "steps 3–6 await review" line is removed.
- **evidence.md** — the traceability matrix gains a DB-002 reference on every affected requirement row: `xmin` concurrency for REQ-011, the quantity CHECK for REQ-013, the product foreign key for REQ-015.
- **status.md** — the next milestone becomes DD-001; the risk "plan revision 1 not yet reviewed" is replaced by the new one, DEC-016.
- **decisions.md** — DEC-013, DEC-014 and DEC-015 recorded as decided, each with its options table and rationale.

Each edit is labelled "Allowed by auto mode classifier" on screen.



```
L Updated work-items\WI-002\plan.md (+3 -3)
67
68 ## Approval / sign-off
69
70 -- **Review status:** submitted for review
71 -- **Approval source:** steps 1-2 performed on the user's direct request of 2026-09-18; steps 3-6 await review of this revision
72 -- **Approved revision:** none yet
73 -- **Review status:** approved
74 -- **Approval source:** user message 2026-09-18: 'approved the plan, let move on to DB design' (steps 1-2 had already been done on the user's direct request of the same day)
75 -- **Approved revision:** revision 1, 2026-09-18
L Updated work-items\WI-002\status.md (+7 -0)
1 # Production Order Create/Edit (Screen A) - Status Report
2
3 As of 2026-09-18. Work item state: awaiting-plan-review.
4 As of 2026-09-18. Work item state: in-progress.
5 ## Overall status
Jump to bottom (ctrl+End) ↓
```

Video 01:07. Approval and traceability written into the work item.

Three decisions taken, one handed back

ID	DECIDED BY CLAUDE, BECAUSE IT DOES NOT CHANGE WHAT THE USER SEES
DEC-013	Order numbers come from a per-year counter table. A year's 100,000th order is rejected rather than changing the user's chosen format.
DEC-014	Use PostgreSQL's <code>xmin</code> to detect that someone else changed an order.
DEC-015	Status is stored as text with a CHECK on the four allowed values.

ONE OPEN ITEM — DEC-016

"Today the app and the migrations both connect to the database as the owner account. The database rules call for a separate app login with only the permissions it needs. Changing that means editing WI-001's Compose setup, so I recommend doing it before any deployment beyond your machine. **It doesn't block the design.**"

```
- A mapping from each column to a proposed API field, for DD-001 to confirm.
- Migration impact: it only adds tables. The doc includes the rollback command and notes that rolling back after use deletes all order data.

New decisions: I decided three technical points myself because they don't change how the screen behaves (details in decisions.md):
- DEC-013: order numbers come from the per-year counter table. A year's 100,000th order is rejected rather than changing your chosen format.
- DEC-014: use xmin to detect that someone else changed an order.
- DEC-015: status is stored as text with a check on the four allowed values.

One open item, DEC-016: today the app and the migrations both connect to the database as the owner account. The database rules call for a separate app login with only the permissions it needs. Changing that means editing
WI-001's Compose setup, so I recommend doing it before any deployment beyond your machine. It doesn't block the design.

status.md, evidence.md (traceability now includes DB-002), the database README, and DD-001's open-questions table are updated to match.

Next is step 5: the detailed design (DD-001) plus a rendered mockup. Publishing the mockup as a private artifact will need your go-ahead when I get there.

Brewed for 2m 15s - done 11:00 AM
```

Video 01:24. The end of the turn 1 summary — 2 m 15 s of agent work.

The question that changed the session

THE PROMPT

Are there any open questions left?

Claude's summary had named one open item and implied that was the list. Instead of accepting it, the user asked the agent to go and check.

WHAT CLAUDE DID

Rather than answer from memory, it searched its own artifacts — `grep "\|open|proposed|unresolved"` across the work item and the design documents.

The same move as in the basic-design session, and it paid off the same way.

```
- DEC-014: use xmin to detect that someone else changed an order.  
- DEC-015: status is stored as text with a check on the four allowed values.  
  
One open item, DEC-016: today the app and the migrations both connect to the database as the owner account. The database rules call for a separate app login with only the permi:  
WI-001's Compose setup, so I recommend doing it before any deployment beyond your machine. It doesn't block the design.  
  
status.md, evidence.md (traceability now includes DB-002), the database README, and BD-001's open-questions table are updated to match.  
  
Next is step 5: the detailed design (DD-001) plus a rendered mockup. Publishing the mockup as a private artifact will need your go-ahead when I get there.  
  
* Brewed for 2m 15s · done 11:06 AM  
are there any open questions left ?  
  
* Doing.. (3s · thinking)  
I
```

Video 01:38. The prompt is sent and Claude greps its own artifacts.

Two known — and three the agent had quietly decided

ITEM

- 1 DEC-016, splitting the database logins — already raised
- 2 CSRF protection — parked for DD-001 to settle
- 3 **Which plant timezone.** "You chose one configured plant timezone, but not which one."
- 4 **Cancel with unsaved changes.** BD-001 leaves the page without asking; should it confirm?
- 5 **Sample products.** Five placeholder products are seeded — swap them for something more realistic?

WHY IT MATTERS

Items 3 to 5 were gaps the agent had closed on its own without recording them as questions. Being asked to re-check surfaced them **before** the detailed design was built on top of them.

```
1. DEC-016, splitting the database logins. Today the app and the migrations both connect as the database owner. My recommendation is to keep that for the local demo and split them before any deployment beyond your machine. It needs your call eventually, but it doesn't block the detailed design.
2. CSRF protection (DD-001, open questions). This is a technical point that the detailed design (DD-001) is meant to settle. No extra token is needed (same-origin cookie with SameSite=Lax and no CORS). I'll confirm or revisit that while writing DD-001.

Rechecking just now, I also found three gaps that aren't recorded as open questions. I filled them in myself, and you should confirm them:
3. Which plant timezone. You chose one configured plant timezone (DEC-011), but not which one. It decides what "today" means for due dates and which year goes into PD-YYYY-... It needs a concrete value such as Asia/Ho_Chi_Minh or Asia/Tokyo.
4. Cancel with unsaved changes. BD-001 says Cancel just leaves the page without asking. A common alternative is to ask "Discard your changes?" when the form has been edited.
5. Sample products. DD-002 seeds five made-up products (P-1001 Steel bracket - P-1005 Hydraulic valve). They're placeholder demo data. Swap them if you want something more realistic for the videos.

Items 3 and 4 change what users see, so I'd like your answers before DD-001 builds on them. Tell me the timezone and whether Cancel should ask to confirm, and I'll record them and go on to DD-001.
```

Video 01:53. "I filled them in myself, and you should confirm them."

Four questions, one form

The user starts typing a direct answer to the two questions Claude asked for — "Asia/Ho_Chi_Minh, and yes, confirm on cancel" — then deletes it and asks for all five instead: **"Let's answer all of them"**.

TIMEZONEDEC-017

Which zone sets "today" and the year in PO-YYYY?

Ho_Chi_Minh · Tokyo · UTC

CANCELDEC-018

Cancel after editing the form — then what?

Confirm if edited · leave without asking

PRODUCTSDEC-019

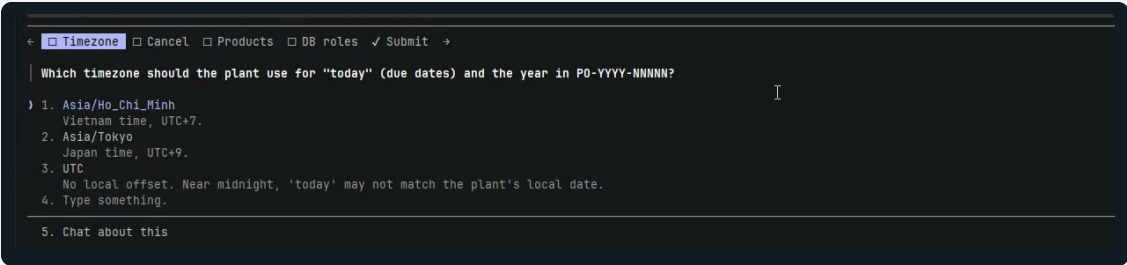
What should the demo seed as products?

5 · ~15 · free text: 30

DB ROLESDEC-016

How should database logins be handled?

split before non-local · split now



Video 02:16. Tab 1 of the form.



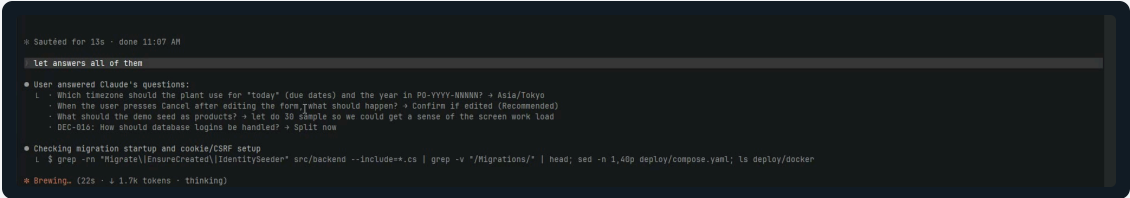
Video 02:35. A free-text answer: "Let's do 30 samples so we can get a sense of the screen's workload".

Every answer lands in the files, not just the chat

ID	QUESTION	RECORDED OUTCOME
DEC-016	Database logins	Split now: the owner login only runs migrations; the backend runs as a restricted <code>pmai_app</code> login, in local Compose too
DEC-017	Plant timezone	<code>Asia/Tokyo</code> , kept as a config value
DEC-018	Cancel with unsaved changes	Ask "Discard your changes?" only if the form was edited — becomes REQ-019
DEC-019	Demo product seed data	30 sample products, <code>P-1001</code> – <code>P-1030</code>
DEC-020	CSRF protection	Claude's own technical call: no extra token — <code>SameSite=Lax</code> cookie, no CORS, JSON-only bodies

CHECKED BEFORE WRITTEN

Before recording anything, Claude re-read the code the answers affect — the migration startup path, the cookie setup in `DependencyInjection.cs` and `deploy/compose.yaml` — then confirmed the cookie configuration matches what ADR-0002 assumed.



Video 02:56. The four answers, as Claude Code recorded them.

The choice is accepted — and its cost is reported

PLAN SCOPE

"Because the login split is happening now, the implementation plan (revision 2) will need to change WI-001's Docker Compose files: a new required runtime password in `.env.example` with no default; a second connection string; a script that grants the runtime login's permissions; and the migration command in `ai/project.md` updated to run as the owner login."

"That plan revision will come to you for review before any of it is built."

Updated files: `decisions.md`, `brief.md` (REQ-019 added), BD-001 (now revision 3), DB-002, ADR-0002, DB-001, `plan.md`, `status.md` and `evidence.md`.

```
later* note. You can overrule it.

Updated files: decisions.md, brief.md (REQ-019 added), BD-001 (now revision 3), DB-002, ADR-0002, DB-001, plan.md, status.md and evidence.md.

Plan scope: because the login split is happening now, the implementation plan (revision 2) will need to change WI-001's Docker Compose files:
- a new required runtime password in .env.example, with no default;
- a second connection string;
- a script that grants the runtime login's permissions;
- updating the migration command in ai/project.md to run as the owner login.

That plan revision will come to you for review before any of it is built.

Next is step 5: the detailed design (DD-001) plus a rendered mockup. I'll ask before publishing the mockup.

» Sautted for 1m 51s - done 11:12 AM
```

Video 03:23. "Nothing in Screen A's design is open now."

OUTPUT

Every decision from DEC-001 to DEC-020 is decided. The design phase's remaining work is DD-001 and a rendered mockup — and Claude states it will ask before publishing that mockup.

The open-decisions table, all resolved

The user reads `decisions.md`, then opens DB-002 and scrolls to the table the document opened with — the one that listed what was still undecided when the schema was written.

DECISION	STATUS
Per-year order sequence	decided — DEC-013
Past 99,999 orders in one year	decided — DEC-013
Optimistic-concurrency token	decided — DEC-014
Status storage	decided — DEC-015
Split migration-owner and runtime roles	decided — split now (DEC-016, user)
Demo product seed size	decided — 30 (DEC-019, user)

The next prompt, "Move on to the detailed design", is typed but not sent. The recording ends there.

src/backend/ProductionManagementAI.Api drops the three tables. Nothing outside them references them.

- Operational note: the generated `order_number` column is defined with EF Core `HasComputedColumnSql(..., stored: true)`. The expression uses explicit `::text` casts because PostgreSQL requires generation expressions to be immutable.

Open decisions

Decision	Options	Recommendation	Status
How to generate the per-year order sequence	per-year counter table with upsert; <code>MAX(order_seq)+1</code> with retry; one PostgreSQL sequence per year	Counter table	decided — see <code>work-items/WI-002/decisions.md</code> DEC-013
What happens past 99,999 orders in one year	reject the create; widen to 6 digits	Reject (CHECK constraint)	decided — DEC-013
Optimistic-concurrency token	PostgreSQL <code>xmin</code> ; explicit <code>version</code> column	<code>xmin</code>	decided — DEC-014
Status storage	<code>varchar</code> + CHECK; PostgreSQL enum type; smallint code	<code>varchar</code> + CHECK	decided — DEC-015
Split migration-owner and restricted runtime DB roles	split now; keep the single owner role for the local demo	Split before any shared/non-local deployment	decided — split now (DEC-016, user)
Demo product seed size	5; ~15; 30	5	decided — 30 (DEC-019, user)

Video 03:52. DB-002's open decisions, every row now carrying a decision ID.

Six things to notice

01

The gate is a file, not a memory

The previous session stopped at plan approval. This one opens by writing that approval into `plan.md` before any design work starts.

04

The human can overrule

On the database-login split Claude recommended waiting. The user said split now — and the agent recorded the choice and the work it creates.

02

Read the code, not just the docs

Naming convention, key types and migrations were read from `AppDbContext` before the schema was written, so DB-002 describes what exists.

05

Free text is a first-class answer

"Let's do 30 samples so we can get a sense of the screen's workload" became DEC-019 and a concrete `P-1001 – P-1030` seed.

03

Asking again finds more

"Are there any open questions left?" made the agent search its own artifacts and surface three gaps it had silently closed.

06

One answer can grow the spec

The Cancel decision did not just edit a document: it added REQ-019 to the brief and a dialog with focus and Escape behaviour to BD-001.

Design phase, second half unlocked

FILE	STATE AFTER THE SESSION
0002-production-order-schema.md	Created; tables, constraints, grants, API mapping
decisions.md	DEC-001 to DEC-020 decided
brief.md	REQ-019 added
BD-001-...-create-edit.md	Revision 3
plan.md	Revision 1 approved; revision 2 scope known
status.md	Green; next action is DD-001
evidence.md	Traceability now reaches the schema

Sources: the session recording (`ScreenA_DB.mp4`), its transcript, and the committed files in `work-items/WI-002/` and `docs/en/database/` .

NEXT IN THE SERIES

- **Detailed design (DD-001)** — API contract, function and screen-processing designs, and a rendered mockup. Publishing the mockup needs the user's go-ahead.
- **Implementation** — including the restricted runtime login this session decided on, and code and tests at every level.

CARRIED FORWARD

DEC-016 turns into real work: a new password in `.env.example` , a second connection string, a grant script, and a migration command that runs as the owner.